

EXPERIMENT NO. 7:

Write a program to implement Logistic Regression (LR) algorithm in Python

Aim:

To implement the Logistic Regression (LR) algorithm in Python and classify the given dataset.

Procedure:

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Create the Logistic Regression model.
5. Train the model using the training dataset.
6. Predict the class labels for the test dataset.
7. Calculate the accuracy of the model.
8. Display the predicted class and accuracy.

Program:

Logistic Regression Algorithm

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

Load Iris Dataset

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

Split dataset into training and testing sets

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.3, random_state=42
```

```
)
```

```
# Create Logistic Regression model
model = LogisticRegression(max_iter=200)

# Train the model
model.fit(X_train, y_train)

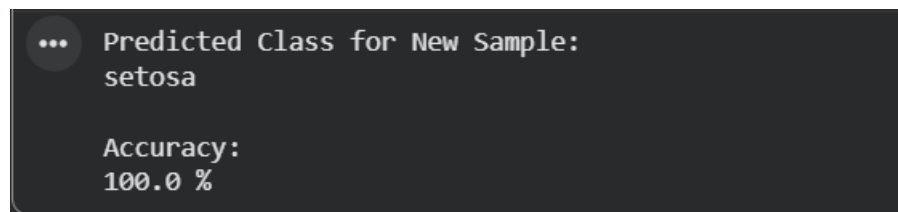
# Predict test data
y_pred = model.predict(X_test)

# Predict a new sample
new_sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(new_sample)

print("Predicted Class for New Sample:")
print(iris.target_names[prediction][0])

# Calculate Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:")
print(round(accuracy * 100, 2), "%")
```

Output:

A terminal window with a dark background and light gray text. The first line shows a prompt '...' followed by the text 'Predicted Class for New Sample:' and 'setosa' on the next line. The second line shows 'Accuracy:' followed by '100.0 %' on the next line.

```
... Predicted Class for New Sample:
    setosa

    Accuracy:
    100.0 %
```

Accuracy:

100.0 %

Result:

The Logistic Regression (LR) algorithm was successfully implemented in Python using the Iris dataset. The model correctly classified the new sample as Setosa and achieved high classification accuracy.